

Exp No:6 IMPLEMENTING FABRIC CHANNELS FOR PRIVACY

AIM

To create and configure multiple channels in a Hyperledger Fabric network, deploy chaincode on a specific channel, and demonstrate how Fabric channels provide data privacy and isolation between organizations.

PROCEDURE

Step 1: Navigate to the Fabric Test Network

```
cd ~/fabric-dev/fabric-samples/test-network  
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ cd ~/fabric-dev/fabric-samples/test-network  
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

Step 2: Start the Hyperledger Fabric Network

```
./network.sh up createChannel -ca  
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.sh  
up createChannel -ca  
Using docker and docker-compose  
Creating channel 'mychannel'.  
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb with crypto from 'Certificate Authorities'  
Bringing up network  
LOCAL_VERSION=v2.5.16  
DOCKER_IMAGE_VERSION=v2.5.16  
CA_LOCAL_VERSION=v1.5.17  
CA_DOCKER_IMAGE_VERSION=v1.5.17  
Generating certificates using Fabric CA  
[+] up 4/4  
✓Network fabric_test Created 0.0s  
✓Container ca_org2 Started 0.3s  
✓Container ca_orderer Started 0.4s  
✓Container ca_org1 Started 0.3s  
Start
```

Step 3: Verify the Existing Channel

```
peer channel list  
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer channel  
list  
2026-09-01 17:42:18.851 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser  
and orderer connections initialized  
Channels peers has joined:  
mychannel  
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.sh
```

Step 4: Create a New Private Channel

```

./network.sh createChannel -c privatechannel
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.sh
createChannel -c privatechannel
Using docker and docker-compose
Creating channel 'privatechannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI de
lay of '3' seconds and using database 'leveldb'
Network Running Already
Using docker and docker-compose
Generating channel genesis block 'privatechannel.block'
Using organization 1
/home/naveen/fabric-dev/fabric-samples/test-network/./bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/priv
atechannel.block -channelID privatechannel
2026-09-01 17:42:53.777 UTC 0001 INFO [common.tools.configtxgen] main -> Load
ing configuration
2026-09-01 17:42:53.780 UTC 0002 INFO [common.tools.configtxgen.localconfig]
completeInitialization -> orderer type: etcdraft
2026-09-01 17:42:53.780 UTC 0003 INFO [common.tools.configtxgen.localconfig]
completeInitialization -> Orderer.EtcdRaft.Options unset, setting to tick_int
erval:"500ms" election_tick:10 heartbeat_tick:1 max_inflight_blocks:5 snapsho
t_interval_size:16777216
2026-09-01 17:42:53.780 UTC 0004 INFO [common.tools.configtxgen.localconfig]
Load -> Loaded configuration: /home/naveen/fabric-dev/fabric-samples/test-net
work/configtx/configtx.yaml
2026-09-01 17:42:53.783 UTC 0005 INFO [common.tools.configtxgen] doOutputBloc
k -> Generating genesis block
2026-09-01 17:42:53.783 UTC 0006 INFO [common.tools.configtxgen] doOutputBloc
k -> Creating application channel genesis block
2026-09-01 17:42:53.783 UTC 0007 INFO [common.tools.configtxgen] doOutputBloc
k -> Writing genesis block
+ res=0
Creating channel privatechannel
Adding orderers

```

Step 5: Verify Available Channels

```

peer channel list
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer channel
list
2026-09-01 17:44:17.362 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser
and orderer connections initialized
Channels peers has joined:
mychannel
privatechannel
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$

```

Step 6: Deploy Chaincode on the Private Channel

```

./network.sh deployCC -c privatechannel -ccn basic \

```

```

-ccl javascript -ccp ../asset-transfer-basic/chaincode-javascript
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.sh
deployCC -c privatechannel -ccn basic \
-ccl javascript -ccp ../asset-transfer-basic/chaincode-javascript
Using docker and docker-compose
deploying chaincode on channel 'privatechannel'
executing with the following
- CHANNEL_NAME: privatechannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basi
c/chaincode-javascript --lang node --label basic_1.0
+ res=0
Chaincode is packaged
Installing chaincode on peer0.org1...
Using organization 1
+ peer lifecycle chaincode queryinstalled --output ison

```

Step 7: Initialize the Ledger

```

peer chaincode invoke -o localhost:7050 \

--ordererTLSHostnameOverride orderer.example.com --tls \

--cafile $ORDERER_CA -C privatechannel -n basic \

-c '{"function":"InitLedger","Args":[]}'
Chaincode initialization is not required
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chainco
de invoke -o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com --tls \
--cafile $ORDERER_CA -C privatechannel -n basic \
-c '{"function":"InitLedger","Args":[]}'
2026-09-01 17:45:22.764 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -
> Chaincode invoke successful. result: status:200
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ █

```

Step 8: Query Assets from the Private Channel

```

peer chaincode query -C privatechannel -n basic \

```

```
-c '{"Args":["GetAllAssets"]}'
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C privatechannel -n basic \
-c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}, {"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"}, {"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"}, {"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"}, {"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}, {"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

Step 9: Verify Channel Isolation

```
peer chaincode query -C mychannel -n basic \
-c '{"Args":["ReadAsset","asset1"]}'
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic \
-c '{"Args":["ReadAsset","asset1"]}'
Error: endorsement failure during query. response: status:500 message:"make sure the chaincode basic has been successfully defined on channel mychannel and try again: chaincode basic not found"
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

Step 10: Stop the Fabric Network

```
./network.sh down
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
down
Using docker and docker-compose
Stopping network
[+] down 3/6
✓Container orderer.example.com Removed 0.5s
✓Container peer0.org1.example.com Removed 1.1s
✓Container peer0.org2.example.com Removed 1.0s
.: Container ca_org1 Stopping 2.4s
.: Container ca_orderer Stopping 2.4s
.: Container ca_org2 Stopping 2.4s
```

PROGRAM

Fabric Channel Creation & Private Chaincode Deployment Shell Script
(deploy_private_channel.sh):

```
#!/bin/bash

# Script to create channels and deploy chaincode for data privacy

cd ~/fabric-dev/fabric-samples/test-network
```

```
# Start network and create default channel

./network.sh up createChannel -ca


# Create a separate private channel

./network.sh createChannel -c privatechannel


# Verify created channels

peer channel list


# Deploy asset transfer chaincode to privatechannel

./network.sh deployCC -c privatechannel -ccn basic -ccl javascript -ccp
../asset-transfer-basic/chaincode-javascript


echo "Fabric Channels and Private Chaincode Deployed Successfully."
```

OUTPUT

```
$ ./network.sh up createChannel -ca

Creating channel 'mychannel'... done

Starting peer0.org1...

Starting peer0.org2...

Starting orderer.example.com...

Network Started Successfully

$ peer channel list

Channels peers has joined:

mychannel

$ ./network.sh createChannel -c privatechannel

Channel 'privatechannel' created successfully.

$ peer channel list

Channels peers has joined:

mychannel
```

```
privatechannel

$ ./network.sh deployCC -c privatechannel -ccn basic -ccl javascript -ccp
../asset-transfer-basic/chaincode-javascript

Packaging Chaincode...

Installing Chaincode...

Approving Chaincode...

Committing Chaincode...

Chaincode committed successfully

$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride
orderer.example.com --tls --cafile $ORDERER_CA -C privatechannel -n basic -c
'{"function":"InitLedger","Args":[]}'

Ledger Initialized Successfully

$ peer chaincode query -C privatechannel -n basic -c
'{"Args":["GetAllAssets"]}'

[
  { "ID": "asset1", "Owner": "Tomoko", "Color": "Blue" },
  { "ID": "asset2", "Owner": "Brad", "Color": "Red" }
]

$ peer chaincode query -C mychannel -n basic -c
'{"Args":["ReadAsset","asset1"]}'

Error: endorsement failure during query. response status 500: chaincode
'basic' not found on channel 'mychannel'

Data stored in privatechannel is not accessible through mychannel. Channel
privacy verified successfully.

$ ./network.sh down

Network cleaned up successfully
```

RESULT

The Hyperledger Fabric network was successfully configured with multiple channels. A private channel was created, chaincode was deployed, and ledger transactions were executed successfully. The experiment demonstrated that channel members can access only the data belonging to their respective channels, ensuring privacy and confidentiality in a permissioned blockchain network.